

CONTACT

Accra, Ghana
wilfredasumboya@gmail.com
+233 55 123 4567
linkedin.com/in/wilfred-asumboya
github.com/wilfredasumboya
asumboya-folio.lovable.app

TECHNICAL STACK

Languages

Python, TypeScript, C/C++, SQL, MATLAB

ML / DL

PyTorch, TensorFlow, scikit-learn, HuggingFace, XGBoost

AI Systems

RAG, LLM fine-tuning, multi-task learning, SHAP / explainability, vector DBs

Data

NumPy, Pandas, OpenCV, MONAI, DICOM, nibabel

Backend

Django, FastAPI, Node.js, REST, PostgreSQL, Supabase

MLOps

Docker, Git, W&B, Hugging Face Hub, CI/CD

Domains

Medical imaging, computational biology, digital mental health

EDUCATION

BSc. Biomedical Engineering

University of Ghana · Level 300
Exp. Jan 2028
Dean's Honour List (L100–L300) · UGAANA Scholar

WASSCE — General Science

Notre Dame Seminary SHS
2023
Aggregate 7 · NSMQ 2022
National 1/8 Finalist

LANGUAGES

English — Native

INTERESTS

Biomedical AI · Computational imaging · LLMs for health · Low-cost medical devices · STEM education · Football

PROFILE

AI/ML engineer building deployable systems at the intersection of deep learning and biomedicine. I ship multi-task models, RAG pipelines, and explainable AI for clinical decision support — from research notebook to production API. Currently leading AI at INNOGEN while studying Biomedical Engineering at the University of Ghana (Dean's List, L100–L300).

EXPERIENCE

AI/ML Engineer (Lead) / Jan 2025 – Present

INNOGEN · Accra, Ghana

- Lead a 4-engineer AI team shipping multi-task PyTorch models for diabetes-complication risk prediction with SHAP-based clinician explanations.
- Designed and deployed a Django + RAG pipeline (LangChain, pgvector) serving contextual medical guidance over 12k+ curated documents.
- Built CI/CD for model retraining and evaluation; cut iteration time from days to under 2 hours.

Research Assistant — AI in Medical Imaging / Aug 2024 – Present

Institute of Applied Science & Tech. (IAST), University of Ghana

- Authored DeepMammo: a CNN ensemble for breast-lesion classification on full-field digital mammograms (AUC 0.93 on internal hold-out).
- Developed an nnU-Net airway-segmentation pipeline for low-dose CT, optimised for sub-Saharan clinical data scarcity.
- Prototyping LLM-driven triage agents for digital mental-health screening.

Undergraduate Researcher / 2023 – Present

Dept. of Biomedical Engineering, University of Ghana

- Fourier Ptychography microscopy reconstruction in Python — gigapixel imaging on consumer hardware.
- Neutron Activation Analysis of Fort Amsterdam artefacts — statistical modelling of trace-element signatures.

SELECTED PROJECTS

Nexus Intellisense — RAG agent + Django backend for medical literature search; hybrid BM25 + dense retrieval, streaming LLM responses.

DeepMammo — PyTorch ensemble (EfficientNet + ConvNeXt) for mammography classification with Grad-CAM and SHAP overlays.

Multi-Label Diabetes Risk — Explainable multi-task model predicting 5 diabetes complications jointly; outperforms single-task baselines by 8% F1.

MindMesh — Voice-first digital mental-health companion using Whisper + fine-tuned Llama for low-bandwidth deployment.

EquiBase — Open dataset tooling for African biomedical imaging — DICOM normalisation, de-identification, and federated splits.

RESEARCH OUTPUT

- Multi-Task Explainable Framework for Diabetes Complications (in prep, 2026).
- DeepMammo: Ensemble Learning for Mammographic Lesion Classification (preprint).
- Airway Segmentation in Low-Dose CT for Resource-Limited Settings (ongoing).
- Fourier Ptychography on Commodity Optics (completed report, 2024).

AWARDS

- Dean's Honour List — University of Ghana (L100–L300)
- UGAANA Scholarship — University of Ghana Alumni
- Harvard Health Systems Hackathon — Participant (2025)
- NSMQ Upper East Regional Winner (2022) · National 1/8 Finalist